## *"Decoding Diabetic Peripheral Neuropathy Heterogeneity"*

<table>
<tr><td>Project Title</td><td>Hybrid ML-Based System for DPN Patient Stratification Using a Derived Personalized Linguistic Code for Optimized Treatment Response via Embodied VR Mirror Neuromodulation.</td></tr>
<tr><td>Category</td><td>Translational Medicine (TMED) — Disease Detection and Diagnosis (DIS)</td></tr>
<tr><td>Student</td><td>Mariam Mohamed Ahmed Elelidy</td></tr>
<tr><td>Problem</td><td>- Diabetic peripheral neuropathy (DPN) is a common diabetes complication causing peripheral nerve damage with sensory loss, pain, and motor deficits.<br>1. Prevalence:<br>• Affects ~50% of long-term diabetics, ~130 million globally; ~8–10 million adults in the U.S.<br>2. Prognosis & Stratification Gap:<br>• Heterogeneity impedes personalized diagnosis; conventional tests only detect dysfunction -not interpret- leading to unstable, trial-and-error care.<br>3. Economic Burden:<br>• Annual direct medical costs for DPN are ~$5,000–$25,000 per patient, totaling over $50 billion globally, ~$10 billion in the U.S., and ~$0.5–1 billion in Egypt per year.</td></tr>
<tr><td>Methodology</td><td>1. Data Input<br>• Pathophysiological mechanisms (TNF-α, MDA) capturing inflammation and oxidative stress.<br>• EMG biomarkers (MUNE, MUI, MUAPI) reflecting motor unit loss and dysfunction.<br>• Metadata: demographic variables, durations, clinical severity & pain score.<br><br>2. Model Architecture<br>• Mechanism and metadata encoded via Autoencoder → mechanism latent.<br>• EMG features encoded using 1D-CNN + BiLSTM → EMG latent.<br>➟ Latents fused using Transformer-based cHVAE to generate a Personalized Vector (PV).<br>➢ PV feeds:<br>○ Multi-head MLP classifier → stratification.<br>○ Multi-task MLP regressor → predictions.<br>• Interpretability via SHAP and GNNs-based mechanisms–muscle interaction mapping.<br><br>3. System Output<br>A) Personalized Vector (PV): patient-specific disease signature.<br>B) Patient Stratification Summary:<br>Axis 1: progression dynamics Axis 2: Dominant mechanism Axis 3: Symptom–damage alignment<br>C) Management Risk Window<br>D) Mechanisms–to–Muscle Interaction Map<br>E) Nerve Biological Age<br>F) Predicted Outcome<br>G) Pathophysiological Fragmentation Index (PFI)<br>H) VR-Guided Intervention (Phase II): future behavioral neuromodulation module.</td></tr>
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## Results & Analysis

Table 2: *Overall Executive performance*

| Block | Metrics (Test) |
|---|---|
| **Discrimination** | Accuracy/Precision/Recall ≈ **0.96–0.98**, F1-macro/ROC-AUC ≈ **0.97–0.99** |
| **Error** | MAE = **0.037**, RMSE = **0.061** |
| **Agreement** | $R^2$ = **0.95**, Pearson $r$ = **0.96**, Spearman $\rho$ = **0.98** |
| **Calibration** | ECE ≈ **0.002–0.006**, MCE ≈ **0.09–0.14** |
| **Robustness** | ΔAccuracy (SNR stress) < **3%** |
| **Uncertainty** | bootstrap CI: **stable 95%** |

## Figures

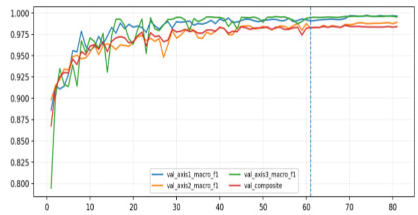


Figure 1: Val Composite

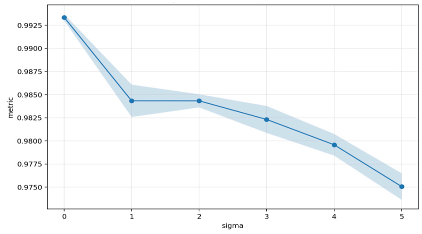


Figure 2: Noise Sensitivity

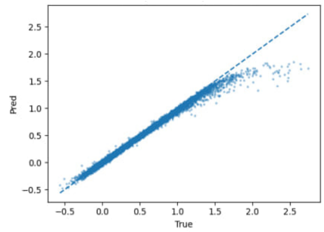


Figure 3: Predicted vs True

## Experiments

### 1) Augmented Generalization Cohort

- **N = 100,000 samples**, Split: **70%** train / **15%** val / **15%** test
- Macro-F1 **0.99** | AUROC **0.997** | Sensitivity **98.74%** | Specificity **99.43%** | MCC **0.982** | Brier **0.014**
- **Generalization** gap (train–test ΔF1): **0.004** | Variance CV-σ (F1): **±0.003**

### 2) External Human Cohort

- **N = 90** (**60** DPN / **30** Control) *"Zero retraining"*
- Accuracy **95.6%** | Macro-F1 **0.954** | AUROC **0.972** | Sensitivity **94.8%** | Specificity **96.7%** | MCC **0.912** | ECE **0.011** | AUROC drop vs internal = **−0.022** *(structural stability preserved)*

***— After confirming external validity, internal structure was tested —***

- Latent cluster separability *(silhouette score)* = **0.67** vs **0.18** (raw clinical grouping)
- **Hidden subtype** separation effect size *(within DPN)* = Cohen's d = **1.31**
- Within-class dispersion reduction after latent encoding = **−42%**
- Intra-DPN progression gradient **correlation** = **r = 0.91**
- Reclassification gain beyond binary clinical label = **+39%**
- Axis monotonicity across oxidative–inflammatory gradients = **100% directional consistency**

### 3) Meta-Anchor & Physiological Stress Alignment

- **r** *(oxidative–inflammatory concordance)* = **0.89** | Cross-marker coherence = **0.92**
- Slope amplification vs pooled literature = **+24%** | Trend concordance gain = **+39%**
- Effect size (model stratification vs severity gradients) = Cohen's d = **1.42**
- Nonlinear fit **$R^2$ = 0.93** | Hazard-direction agreement **= 87%** | Directional inversion **= 0.002%**
- Local slope *(k-region)* amplification factor: **×2.4** *vs* baseline zone

## Impact

Table 2: *Performance estimates derived from internal cross-validation;*

| Key Benefit | Value |
|---|---|
| Reduced Trial-and-Error | ↓ 60% |
| Lower Risk of Complications | ↓ 35% |
| Reduced Treatment Cost | ↓ 45% |
| Personalization Level | ↑ 95–98% |
| Reliable Decision Support | ↑ 92% consistency |
| Early Risk Identification | ↑ 73% |
| Disease Re-Interpretation | Model-driven shift in disease understanding |